

INDIVIDUAL ASSIGNMENT 1

ML System Design Document

Production-oriented requirements, metrics, architecture, and trade-off analysis

Scenario: Human-in-the-loop breast cancer diagnostic support using cell-nucleus measurements. The model prioritizes cases for clinician review; it never issues an autonomous diagnosis.

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Academic note

Operational targets in this report are proposed design thresholds. Prototype results come from the workspace WDBC pipeline and are not evidence of clinical safety or regulatory approval.

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1 Problem definition

1.1 Context and background

A diagnostic laboratory receives fine-needle aspiration measurements from breast masses. Each case contains thirty numeric descriptors of cell nuclei, such as radius, texture, perimeter, area, smoothness, compactness, concavity, symmetry, and related worst/error measurements. The current prototype uses the Wisconsin Diagnostic Breast Cancer (WDBC) structure. The workspace extract has 570 source rows; the validation policy retains 564 rows for modeling and quarantines invalid or duplicate records.

The operational challenge is not to replace the clinician. It is to rank incoming cases by malignant risk, make the evidence traceable, and bring higher-risk cases to a qualified reviewer sooner while preserving a safe manual path whenever data or infrastructure is unreliable.

1.2 Measurable problem statement

Design a production ML decision-support service that assigns a calibrated malignancy-risk score and review priority to every valid case. Within a 12-week pilot, the service should reduce median time to first clinical review by at least 30% for high-risk cases while maintaining malignant-case sensitivity of at least 95%, end-to-end scoring latency below 2 seconds at the 95th percentile, and 100% clinician confirmation before a diagnosis is recorded.

1.3 Current situation

Cases are processed in arrival order, technicians perform manual completeness checks, and clinicians review measurements without a consistent risk-ranking mechanism. Rework is handled through messages and spreadsheets, so rejection reasons, model versions, and review outcomes are difficult to audit. During peaks, potentially malignant cases may wait behind routine cases.

1.4 Why machine learning is appropriate

- The decision boundary depends on interactions among many continuous features; hand-written thresholds lose information.
- A supervised classifier can learn from confirmed labels and produce a probability-like ranking rather than a brittle yes/no rule.
- Performance can be measured offline, monitored in production, and improved through controlled retraining.
- Rules remain essential for validation, privacy, and fail-safe behavior; ML is used only where statistical pattern recognition adds value.

1.5 Stakeholders

Stakeholder	Goal	Primary concerns	Decision / interaction
Clinician / pathologist	Review urgent cases earlier	False negatives, explanation, alert fatigue	Accept, override, or request recollection
Laboratory technician	Submit complete cases once	Clear validation feedback, low rework	Correct quarantined records
Patient	Timely, fair, confidential care	Privacy, bias, accountability	Provides consent under clinical policy
Clinical operations owner	Reduce turnaround time safely	Capacity, adoption, measurable benefit	Sets service levels and pilot scope
Data scientist / ML engineer	Reliable learning system	Drift, leakage, reproducibility	Trains, evaluates, promotes models
Platform / SRE team	Stable and observable service	Uptime, rollback, cost	Operates APIs, pipeline, and alerts
Security / compliance	Lawful and auditable processing	Least privilege, retention, breach risk	Approves controls and access

2 Requirements analysis

2.1 Functional requirements

ID	Requirement	Acceptance evidence
FR-01	Ingest a case with sample ID, 30 numeric measurements, acquisition time, and source metadata.	Schema-valid case receives an immutable ingestion ID.
FR-02	Validate labels, ranges, nulls, duplicate IDs, and extreme-area values; quarantine invalid records.	Reason codes are stored; a batch fails when rejected fraction exceeds 5%.
FR-03	Transform features with statistics fitted only on approved training data.	Training manifest stores scaler parameters and feature order.
FR-04	Return malignancy risk, priority band, model version, timestamp, and explanation summary.	Response contract passes automated schema and unit tests.
FR-05	Require clinician review before any result enters the diagnostic record.	No autonomous diagnosis route exists; every outcome has reviewer identity.
FR-06	Integrate with laboratory information and identity systems through authenticated APIs.	Successful end-to-end test in a de-identified staging environment.
FR-07	Log input fingerprint, output, model version, override, and reason without duplicating raw identifiers.	Audit query reconstructs each decision lineage.
FR-08	Detect distribution, quality, latency, and outcome-performance drift.	Dashboard and alert tests cover each monitored condition.
FR-09	Support shadow, canary, rollback, and champion/challenger releases.	Previous approved version can be restored within 15 minutes.
FR-10	Provide batch backfill and idempotent reruns by logical date.	Rerun updates only its own partition and does not duplicate history.

2.2 Input/output and interaction contract

Element	Specification
Input	case_id (pseudonymous), 30 ordered float features, acquisition timestamp, source device, schema version
Output	risk_score [0,1], priority {urgent, routine, manual-only}, threshold version, model version, quality flags, explanation
User flow	Technician submits -> validation feedback -> clinician worklist ranks valid cases -> clinician confirms/overrides -> outcome is audited
Fallback	On invalid input, missing model, dependency failure, or drift stop: mark manual-only and preserve arrival order
Integration	Laboratory information system, clinician portal, SSO/RBAC, audit store, monitoring, model registry

2.3 Non-functional requirements

Quality attribute	Target	Verification
Performance	p95 API latency <2 s; 100 cases/min sustained; UI update <3 s	Load test with production-like payloads
Scalability	10x daily volume without redesign; stateless inference replicas	Horizontal scaling test; queue-depth alarm
Reliability	99.5% monthly availability; RPO 15 min; RTO 60 min	Failure injection, restore drill, SLO dashboard
Maintainability	Versioned contracts; modular stages; monthly dependency scan; quarterly model review	CI gates, runbook review, ownership map
Security	TLS in transit, encryption at rest, RBAC, secret manager, least privilege	Threat model, access review, penetration test
Observability	100% scored cases traceable; metrics/logs separated from direct identifiers	Lineage query and alert simulation
Usability	Risk and rationale visible without adding >15 seconds to review	Clinician usability test

2.4 Data requirements

Dimension	Requirement and control
Sources	Prototype: versioned WDBC-style CSV. Production: analyzer export plus confirmed pathology outcome from approved systems.
Quality	Schema and units, null/range checks, unique IDs, label allow-list, extreme-value policy, class-distribution checks.
Volume	Prototype: 570 rows and 30 features. Pilot planning assumption: up to 10,000 cases/day with 24 months retained for monitoring.
Privacy	Pseudonymize case IDs, exclude direct identifiers from features/logs, encrypt data, role-based access, retention schedule, audited export.
Representativeness	Report performance by site, device, age band, and other approved cohorts; hold back a temporal/site test set.
Label quality	Use confirmed pathology outcomes; preserve adjudication status and exclude unresolved labels from supervised training.

2.5 Safety, fairness, and governance

The system is a prioritization aid, not a medical device claim. Deployment requires local clinical, legal, privacy, and regulatory review. A model card records intended use, prohibited use, training population, thresholds, known limitations, and subgroup evidence. Overrides are expected and are reviewed for systematic error rather than treated as user failure.

3 Goals and metrics



Figure 1. Goal hierarchy connecting business outcomes to observable system and model behavior.

Level	Goal	Metric	Pilot threshold	Baseline / guardrail
Business	Faster review of high-risk cases	Median time from ingestion to first clinician review	$\geq 30\%$ reduction	Compare with 4-week pre-pilot baseline
Business	Adopted but controlled workflow	% eligible cases scored; clinician override rate	$\geq 90\%$ coverage; investigate $>15\%$ overrides	No automated diagnosis
System	Fast, available service	p95 latency; monthly availability	< 2 s; $\geq 99.5\%$	Manual-only fallback on failure
System	Stable data quality	Rejected row rate; schema failures	$\leq 5\%$; zero silent schema changes	Quarantine rather than impute silently
Model	Do not miss malignant cases	Sensitivity / recall for malignant class	≥ 0.95 on locked test set	Primary model metric
Model	Useful ranking	ROC-AUC; PR-AUC	≥ 0.97 ROC-AUC; tracked PR-AUC	Report confidence intervals
Model	Limit needless escalation	Specificity and precision	≥ 0.85 specificity	Cannot trade away recall floor
Model	Trustworthy risk	Brier score and calibration error	No worse than approved champion	Recalibrate before changing rank model
Monitoring	Detect change early	PSI/KS, missingness, label-lagged recall	Warn PSI >0.10 ; stop review at PSI >0.25	Cohort and site slices required

Metric interpretation

Accuracy is secondary because class imbalance and asymmetric harm can make a high-accuracy model unsafe. The threshold is selected on validation data to satisfy the sensitivity floor, then specificity and workload are optimized. The final threshold is frozen before the locked test set is opened. Business impact is evaluated in a time-bounded silent and assisted pilot, not inferred from ROC-AUC alone.

4 High-level architecture design

Online decision-support path with offline governance

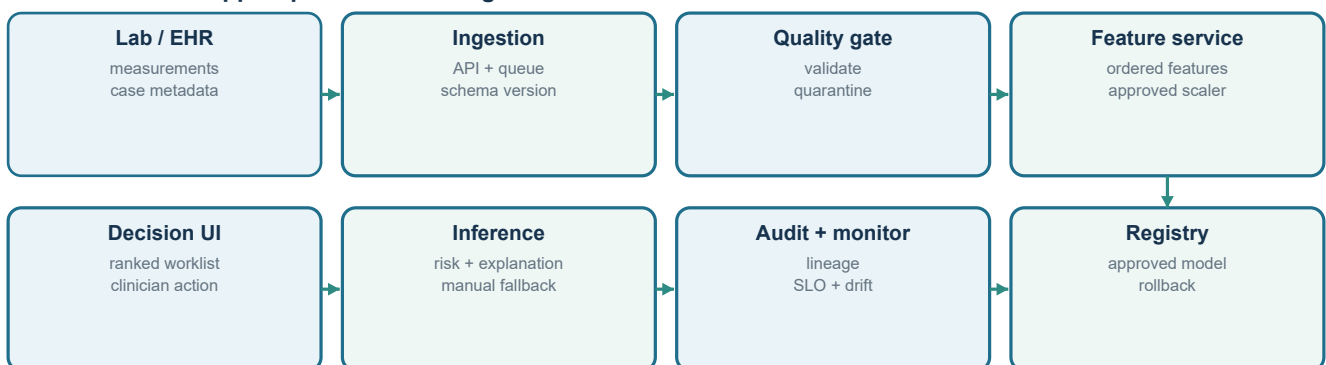


Figure 2. High-level production architecture. All diagnostic decisions remain human-controlled.

4.1 Data flow

1. The laboratory system submits a pseudonymous case and schema version through an authenticated endpoint.
2. Ingestion snapshots the payload and emits an immutable event; validation applies deterministic quality rules.
3. Valid measurements are reordered by a versioned feature contract and transformed with approved training statistics.
4. The inference service loads a registry-approved model, scores the case, and records model and threshold versions.
5. The clinician portal displays priority, risk, quality flags, and a concise explanation; the clinician confirms or overrides.
6. Audit, SLO, quality, drift, and delayed outcome metrics feed operational dashboards and the governed retraining pipeline.

4.2 Offline ML lifecycle

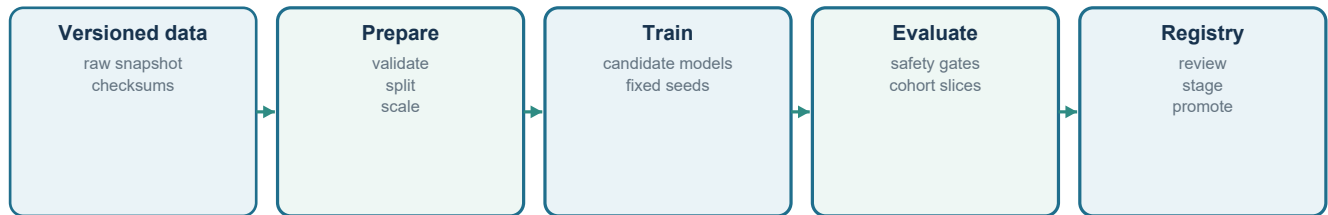


Figure 3. Offline training path. Only an approved registry version can enter inference.

4.3 Component responsibilities and technology choices

Component	Responsibility	Candidate technology	Key control
API / queue	Receive, authenticate, buffer, and deduplicate cases	FastAPI + managed queue	Idempotency key and schema registry
Validation	Apply deterministic quality policy and quarantine	Pandas/Great Expectations	Fail batch above 5% rejection
Feature transformation	Enforce order, types, and scaler version	scikit-learn pipeline	No train/test leakage
Inference	Return score, band, explanation, version	Containerized Python service	Timeout -> manual-only
Orchestration	Run and retry offline stages by logical date	Apache Airflow	Idempotent partitions and bounded retries
Tracking / registry	Store params, metrics, artifacts, approvals	MLflow	Immutable run ID and versioned model
Storage	Raw snapshot, curated data, reports, audit	Object store + relational audit DB	Encryption, retention, access policy
Monitoring	SLO, data quality, drift, outcome metrics	Prometheus/Grafana + drift jobs	Alert routing and runbooks

4.4 Failure behavior

Transient infrastructure errors retry with exponential backoff. Deterministic data errors do not burn retries; they quarantine the case and notify the technician. If the model or registry is unavailable, the portal shows manual-only status and preserves arrival order. A circuit breaker prevents repeated calls to a failing dependency. Every fallback is observable and included in the availability SLO.

5 Trade-off analysis

Trade-off	Option A	Option B	Decision and rationale
Sensitivity vs specificity	Lower threshold finds more malignant cases but increases urgent reviews	Higher threshold reduces workload but risks false negatives	Set a sensitivity floor first; optimize specificity only within that safe region.
Accuracy vs latency	Complex ensembles can improve ranking	Linear models are faster and easier to inspect	Start with regularized logistic regression; promote a complex model only for material, repeatable benefit.
Freshness vs cost	Continuous retraining reacts quickly	Scheduled retraining is cheaper and easier to validate	Monitor continuously; retrain quarterly or on approved drift trigger, never promote automatically.
Privacy vs personalization	More patient/context features may improve calibration	Minimal data reduces privacy and bias exposure	Use only clinically justified features; direct identifiers never enter the model.
Automation vs control	Automatic priority saves time	Manual review preserves accountability	Automate ranking and validation, but require clinician confirmation and preserve override reasons.
Centralization vs resilience	Single central service simplifies governance	Site-local fallback reduces outage impact	Use a governed central model plus cached approved version and manual-only mode at each site.

Residual risks

Risk	Mitigation	Residual decision
Prototype data is small and not site-representative	External and temporal validation; silent pilot; subgroup reporting	No clinical use until local evidence and approvals exist
Label delay masks current performance	Track proxy drift now and outcome metrics when labels mature	Escalate sustained drift to model review board
Automation bias	Display uncertainty and limitations; train users; sample low-risk cases	Monitor overrides and missed-case reviews
Data leakage or train/serve skew	Versioned feature contract; scaler fitted on train only; parity tests	Block promotion on failed lineage or parity gate

6 Delivery roadmap and risk controls

Phase	Weeks	Deliverable	Exit gate
Discovery and governance	1-2	Workflow map, threat model, data dictionary, intended-use statement	Clinical, privacy, and security sign-off
Offline prototype	3-5	Versioned dataset, reproducible baseline, error analysis	Locked test evidence meets candidate thresholds
Shadow integration	6-8	Scores hidden from clinicians; SLO and drift dashboards	No silent data-contract or reliability failures
Assisted pilot	9-12	Limited-site worklist ranking with override	Business benefit and safety review
Controlled scale	After 12	Canary rollout, periodic review, rollback drills	Model review board approval per release

Go/no-go policy

A release is blocked if lineage is incomplete, the locked-test sensitivity floor is missed, subgroup evidence is materially worse without an approved mitigation, calibration regresses, dependency scanning fails, or rollback cannot be demonstrated. Meeting model metrics alone is never sufficient for release.

7 Conclusion and references

The proposed system turns a classification model into a governed clinical workflow: deterministic validation, versioned transformations, controlled inference, human confirmation, complete lineage, observable fallback, and measurable business evaluation. The architecture deliberately favors safety, auditability, and reversible deployment over autonomous decision making.

References

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